



TRACE THE ACE
Transcript Intelligence Challenge

TRACE THE ACE

BASELINE INTELLIGENCE REPORT

From Dataset Exploration to Model Failure Analysis

EDA • Feature Engineering • Baseline Models • Optuna Tuning • Research Directions

MODEL PERFORMANCE (BASELINE)



00:58:12 TUTOR: Let us examine the student response...
00:58:16 STUDENT: I think the answer is forty-five.
00:58:20 TUTOR: Explain why that relationship holds.
00:58:24 STUDENT: Because the angles sum to 180.

Team SkyMap

DATA

PREPROCESSING

FEATURES

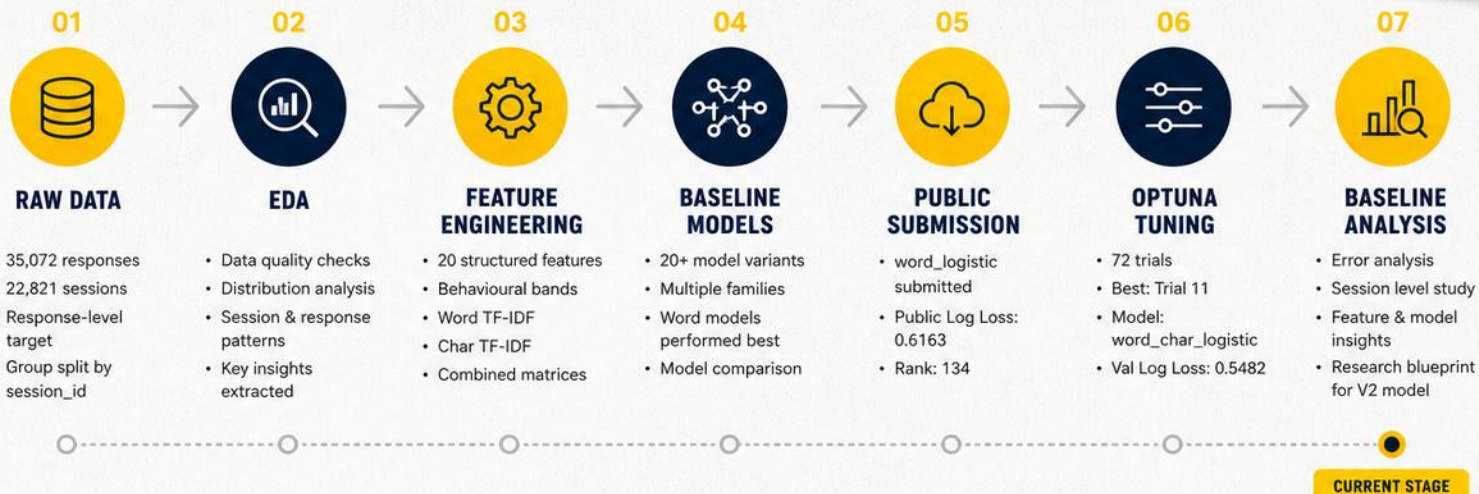
MODELING

EVALUATION

OUR BASELINE JOURNEY

From Raw Data to Research Insights

Our end-to-end baseline pipeline covered data exploration, feature engineering, multiple model families, public submission and hyperparameter tuning. Now we move to deep baseline analysis to understand what the model gets right, what it misses, and why.



OUR NEXT GOAL

Deep baseline analysis to uncover failures, validate hypotheses and design a stronger next-generation model.

“ Better understanding today,
Smarter models tomorrow. ”

DATASET LANDSCAPE

What the main dataset looks like before modeling

35,072

RESPONSE ROWS

Individual tutor responses
(prediction rows)

22,821

SESSIONS (GROUPS)

Group split by session_id
(no leakage across splits)



PREDICTION TASK

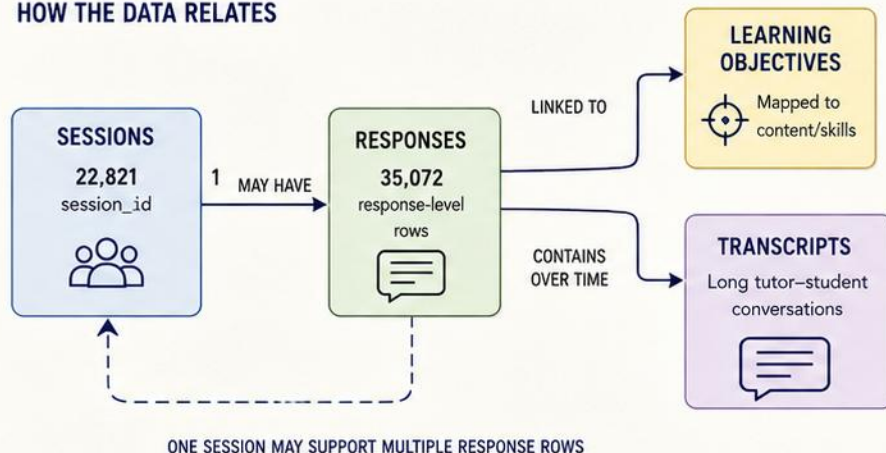
Response-level
prediction of student
correctness (ACE)

~70.25%

POSITIVE RATE

Target is imbalanced
toward positive (correct)

HOW THE DATA RELATES



TARGET DISTRIBUTION (ACE)



- Positive (Correct)
70.25% (24,610)
- Negative (Incorrect)
29.75% (10,462)

Totals may not sum exactly
due to rounding.

TRANSCRIPT SAMPLE (FROM ONE SESSION)

Tutor: Let's try another way. What do we know about triangles?
Student: The angles add up to 180 degrees.
Tutor: Good. If one angle is 90 and another is 45, what is the last angle?
Student: 45.
Tutor: Correct. Why is that?
Student: Because $90 + 45 + 45 = 180$.



KEY TAKEAWAY

The dataset is response-level, but much of the context is session-level.

WHAT EDA REVEALED

Useful signals, structural risks, and early clues

USEFUL SIGNALS

01 Student response depth mattered.

Deeper, more elaborated responses aligned with higher outcomes.

02 Text carried stronger signal than summary statistics.

Raw transcript text outperformed features built from counts and aggregates.

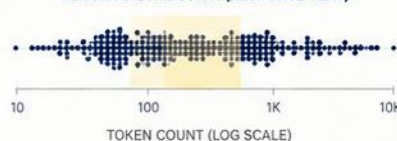
03 Session behavior patterns were informative.

Response timing, ordering, and turn dynamics showed meaningful associations.

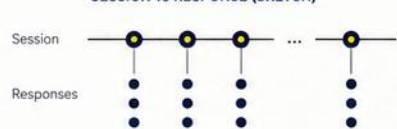
04 Learning objective text was useful.

Objective wording was predictive and complemented response text.

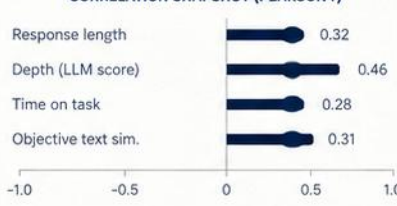
LENGTH DISTRIBUTION (RESPONSE TEXT)



SESSION vs RESPONSE (SKETCH)



CORRELATION SNAPSHOT (PEARSON r)



DATA RISKS

01 Class imbalance.

High-performing outcomes dominate the dataset.

02 Multiple responses inside the same session.

Responses are not independent; leakage risk if split improperly.

03 Very long transcripts.

Heavy-tailed length distribution adds noise and compute cost.

04 Potential information dilution.

Mixing exploratory and off-topic turns may weaken signal.

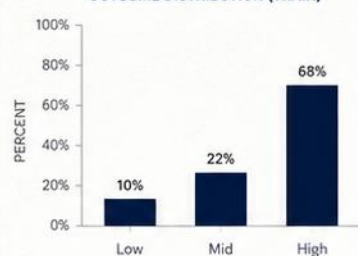
05 Repeated session-level information.

Objectives and context repeat across responses within a session.

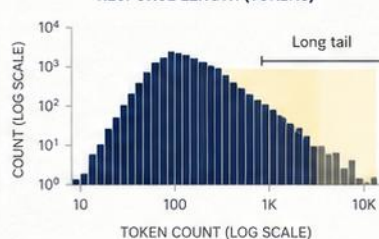
06 Possible local-to-public distribution shift.

Local validation may overestimate generalization to public test set.

OUTCOME DISTRIBUTION (TRAIN)



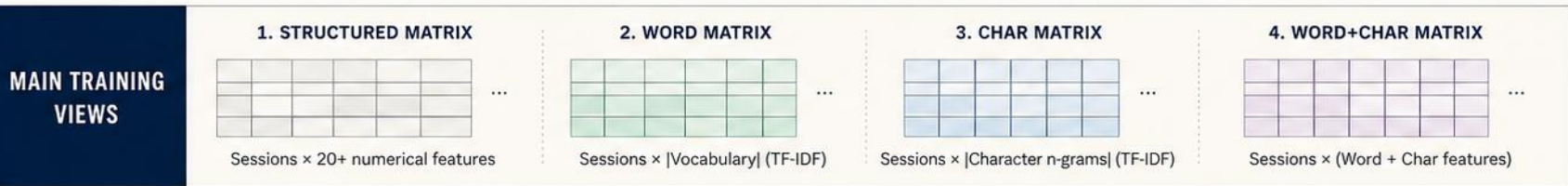
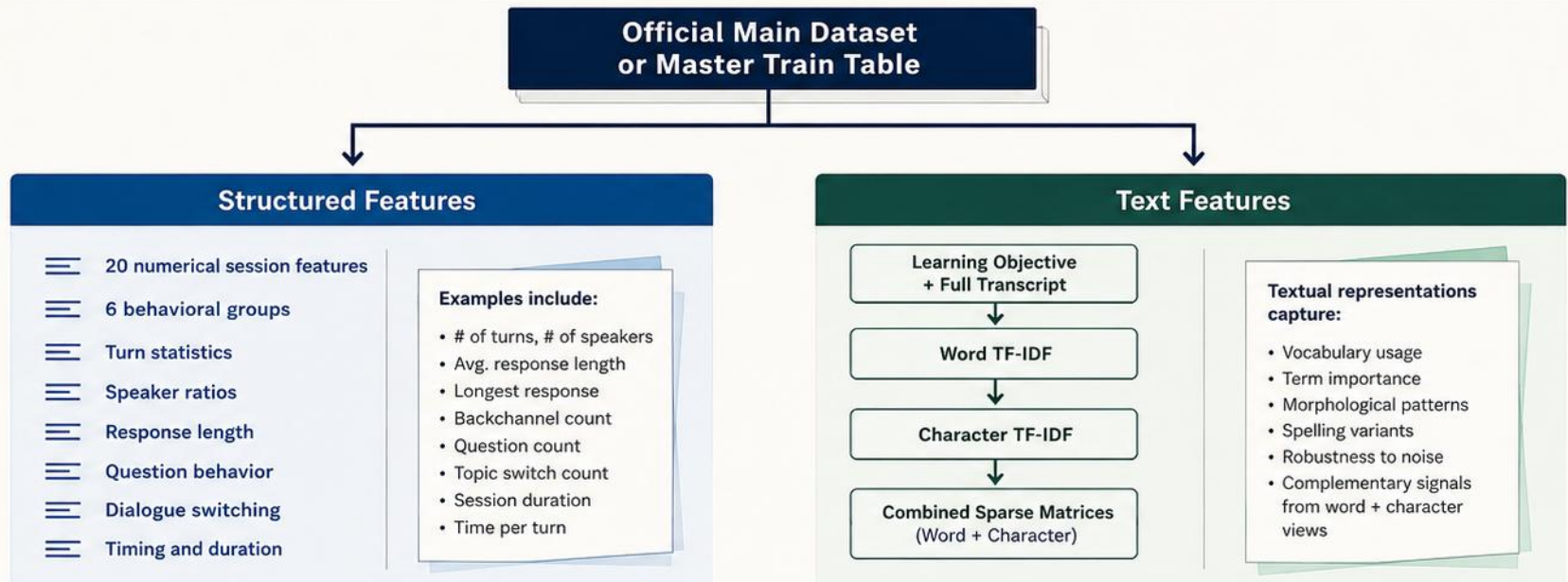
RESPONSE LENGTH (TOKENS)



EDA showed that transcript text is informative, but the problem structure is more complex than a standard flat classification task.

FEATURE SYSTEM WE BUILT

How baseline inputs were transformed into model-ready representations



TAKEAWAY

The baseline relied on a simple, reproducible feature stack before advanced modeling.

WHAT FEATURE ENGINEERING TAUGHT US

Which feature families carried real predictive signal

PREDICTIVE SIGNAL COMPARISONS

DUMMY BASELINE

0.5986

LOG LOSS

>

1

STRUCTURED GAIN
OVER DUMMY IS SMALL

0.0061

LOWER LOG LOSS IS BETTER

BEST STRUCTURED MODEL

0.5924

LOG LOSS

>

2

TEXT GAIN OVER
STRUCTURED IS MUCH LARGER

0.0470

LOWER LOG LOSS IS BETTER

BEST WORD MODEL

0.5455

LOG LOSS

ALL-FEATURES LOGISTIC

0.5508

LOG LOSS

RUNTIME & PERFORMANCE TRADE-OFF

WORD LOGISTIC

6.9 SEC

0.5455

LOG LOSS

74×
SLOWER

ALL FEATURES LOGISTIC

510.2 SEC

0.5508

LOG LOSS

**WORSE
SCORE**

TAKEAWAY: Text carried the strongest predictive signal;
current structured features were useful but weak and mostly session-level.



BASELINE MODEL TOURNAMENT

How the main model families performed

TOP PERFORMERS – SPARSE LINEAR TEXT MODELS DOMINATED

LOG LOSS (↓)

1	word_logistic		0.5455
2	word_char_logistic		0.5504
3	word_sgd		0.5504
4	word_char_sgd		0.5507
5	all_features_logistic		0.5508
6	all_features_sgd		0.5532
7	tfidf_logistic		0.5551
8	tfidf_sgd		0.5560
9	word_logistic_l2		0.5573
10	word_sgd_l2		0.5592

STRUCTURED MODELS – WEAKER

LOG LOSS (↓)

random_forest		0.5924
extra_trees		0.5961
gradient_boosting		0.5987
adaboost		0.6052

POOR OUTLIERS

LOG LOSS (↓)

structured_decision_tree		4.8835
word_bernoulli_nb		5.0658

Lower log loss is better.
Results are on the validation set.

FAMILY SUMMARY

LOGISTIC REGRESSION

Best overall
performer

SGD

Tightly competitive
with Logistic
Regression

NAIVE BAYES

Generally weak;
worst among
baselines

RANDOM FOREST

Moderate
performance

EXTRA TREES

Moderate
performance

GRADIENT BOOSTING

Moderate
performance

ADABOOST

Moderate
performance

DECISION TREE

Very poor
(outlier)

DUMMY PRIOR

Baseline only;
not predictive

Sparse linear text models clearly outperformed tree-based and Naive Bayes baselines.

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OF 12

TRACE
THE
ACE

TEAM SKYMAP

BEST BASELINE VS PUBLIC REALITY

Why the local winner did not generalize publicly

LOCAL VALIDATION

word_logistic

Log Loss **0.5455**

AUROC **0.7036**

Accuracy **0.7313**

THE CRITICAL GAPS

LOG LOSS GAP
+0.0708

(higher is worse)

AUROC DROP
-0.1030

(lower is worse)

PUBLIC SUBMISSION

Log Loss **0.6163**

AUROC **0.6006**

Rank **134**
out of all teams

The model lost both calibration quality (higher log loss) and ranking ability (lower AUROC) when evaluated on the unseen public test set.

STRONG LOCAL VALIDATION DID NOT TRANSLATE TO STRONG PUBLIC GENERALIZATION.

OPTUNA SEARCH

Focused tuning of the strongest sparse-text families

09
OF 12

72

TRIALS

Explored the hyperparameter space

8

PARALLEL CPU WORKERS

Efficient parallel search

90

MINUTE BUDGET

Wall-clock time limit

MODELS CONSIDERED

Word sparse (word n-grams)

Character sparse (char n-grams)

Combined sparse (word + char)

BEST INTERNAL TRIAL

Trial 59 produced the lowest internal-split log loss.

INTERNAL LOG LOSS

0.5709



FINAL SELECTION

Top candidates were rechecked with 3-fold group cross-validation.

FINAL SELECTED CONFIGURATION

Trial 11 | **word_char_logistic**

C	≈ 0.5255
fit_intercept	True
class_weight	None
solver	saga

TOP 5 CANDIDATES - 3-FOLD GROUP OOF LOG LOSS

1

Trial 11

word_char_logistic

0.562109

2

Trial 43

word_char_logistic

0.562177

3

Trial 59

word_char_logistic

0.562196

4

Trial 58

word_char_logistic

0.562804

5

Trial 44

word_char_logistic

0.562862

Final model selected after re-evaluation, prioritizing stability across session-grouped folds.

WHY TUNING DID NOT IMPROVE THE SCORE

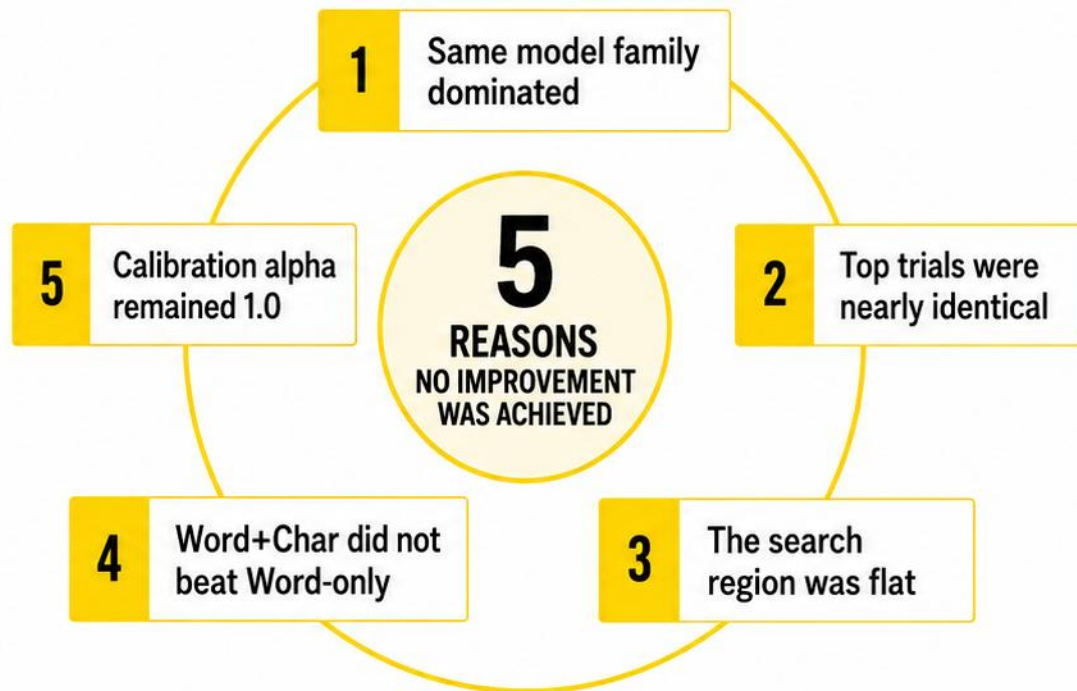
Optimization without new information

WHAT WE OBSERVED

Baseline validation	0.5455
Optuna validation	0.5482
ACTUAL CHANGE	+0.0028 WORSE

WHAT WE HOPED FOR

Expected improvement range +0.002 to +0.006 (roughly) This improvement did not happen.
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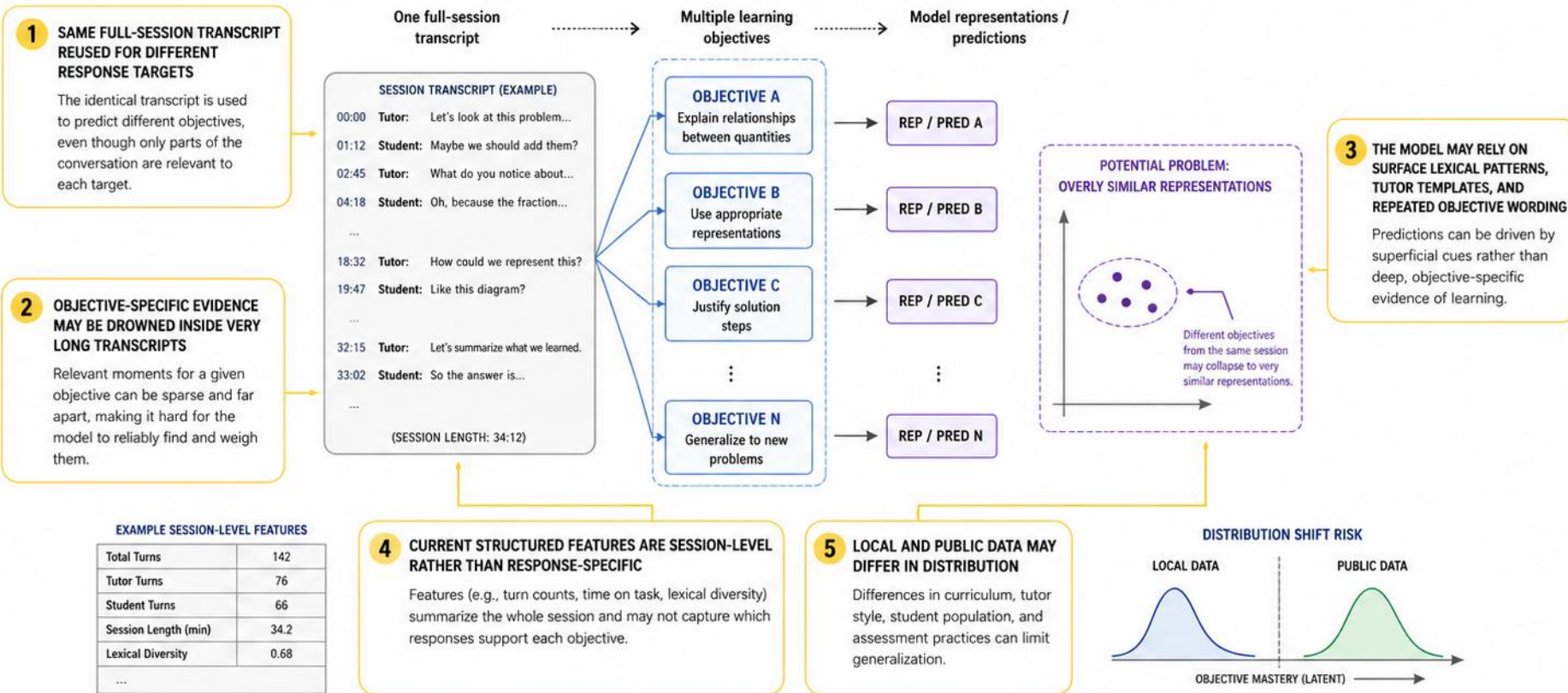


Tuning optimized the existing representation;
it did not create new predictive information.

WHAT THE BASELINE MAY BE GETTING WRONG

The major research questions revealed by the first system

THE FIRST SYSTEM: SESSION → MULTIPLE OBJECTIVES



The baseline is useful, but it likely **under-represents** objective-specific learning evidence.



TRACE THE ACE



TEAM SKYMAP

RESEARCH BLUEPRINT FOR THE 0.59 JOURNEY

What we must investigate before building the next model

01



DATA AND VALIDATION RESEARCH

- ✓ Runtime parity
- ✓ 5-fold group OOF
- ✓ Provider or domain analysis
- ✓ Objective-frequency analysis
- ✓ Train-validation-test drift
- ✓ Duplicate / conflict audit

02



ERROR RESEARCH

- ✓ High-confidence errors
- ✓ Worst sessions
- ✓ Mixed-label sessions
- ✓ Prediction collapse
- ✓ Length-based failures
- ✓ Rare-objective failures

03



REPRESENTATION RESEARCH

- ✓ Objective-conditioned key moments
- ✓ Student-only evidence
- ✓ Tutor-question to student-answer pairs
- ✓ Dialogue progression
- ✓ Tutor and student moves
- ✓ Relevant-turn retrieval

04



V2 MODEL RESEARCH

- ✓ 9 auxiliary files
- ✓ Pretrained transformers
- ✓ Domain-aware experts
- ✓ Semantic pair models
- ✓ Hierarchical transcript models
- ✓ Diverse OOF ensemble



The path to 0.59 is not more tuning. ***It is better task representation.***